

ALGEBRA QUALIFYING EXAMINATION

Winter 2024

PLEASE WRITE YOUR UO ID NUMBER ON YOUR SOLUTIONS but not your name.

There are three parts to this exam:

- Part I is asking for statements of some definitions/theorems. Three questions, 6 points each, give concise but accurate answers.
- Part II is True/False. Five questions, 8 points each, give a sketch proof or counterexample for full credit.
- Part III is longer problems. Three questions, 14 points each, give concise but clear proofs stating any significant results you are using.

ATTEMPT ALL QUESTIONS

Part I. Briefly state each of the following (6 points each):

1. The definition of the determinant of an endomorphism $f : V \rightarrow V$ of a finite-dimensional vector space V .

Solution. Quickest is to let $n := \dim V$. Applying the functor $\bigwedge^n$ to f gives a linear map $\bigwedge^n f : \bigwedge^n V \rightarrow \bigwedge^n V$. Since $\bigwedge^n V$ is one-dimensional, it is multiplication by a scalar. This scalar is the determinant $\det f$.

Or you could pick a basis $v_1, \dots, v_n$ for V . Then let $A = (a_{i,j})_{1 \leq i,j \leq n}$ be the matrix of f defined by $f(v_j) = \sum_{i=1}^n a_{i,j} v_i$. Then

$$\det f = \sum_{g \in S_n} \operatorname{sgn}(g) a_{1,g(1)} a_{2,g(2)} \cdots a_{n,g(n)}.$$

2. Two different but equivalent definitions of an injective module.

Solution. Let R be a ring. An R -module I is injective if any of the following equivalent conditions hold:

- The functor $\operatorname{Hom}(_, I)$ (from the category of R -modules to Abelian groups) is exact.
- Any short exact sequence of R -modules $0 \rightarrow I \rightarrow M \rightarrow N \rightarrow 0$ splits.

- For any injection of R -modules $i : W \rightarrow V$ and any R -module homomorphism $\phi : W \rightarrow I$ there exists a homomorphism $\psi : V \rightarrow I$ such that $\phi = \psi \circ i$.

3. The Krull-Schmidt Theorem for modules of finite length.

Solution. Let R be a ring and V be an R -module of finite length (meaning that V has a composition series). Then V can be written as an internal direct sum $V = V_1 \oplus \cdots \oplus V_n$ of finitely many indecomposable submodules. This decomposition is unique in the sense that if $V = W_1 \oplus \cdots \oplus W_m$ is another such decomposition, then $m = n$ and $V_i \cong W_{g(i)}$ for some $g \in S_n$.

Part II. True or false? If true provide a brief explanation, if false provide a counterexample (8 points each):

1. The dihedral group D_{2024} of order 4048 is nilpotent.

Solution. FALSE. Recall that $Z(D_n)$ is trivial if $n \geq 3$ is odd, and $D_n/Z(D_n) \cong D_{n/2}$ if $n \geq 2$ is even. This follows by a calculation using

$$D_n = \langle a, b \mid a^n = b^2 = 1, bab^{-1} = a^{-1} \rangle = \{a^i b^j \mid i = 0, 1, \dots, n-1, j = 0, 1\};$$

when n is even the central involution is $a^{n/2}$. Hence, the dihedral group D_{253} is a quotient of D_{2024} . Since D_{253} has trivial center it is not nilpotent. Quotients of nilpotent groups are nilpotent, so this implies that D_{2024} is not nilpotent either.

2. Any semisimple $\mathbb{R}$ -algebra of dimension 3 is commutative.

Solution. TRUE. By Artin-Wedderburn, it is a direct sum of matrix algebras over real division algebras. We can't have any matrix algebras that are 2×2 or bigger since the dimension is ≤ 3 . So it is a direct sum of real division algebras. The only finite-dimensional real division algebras of dimension ≤ 3 are $\mathbb{R}$ and $\mathbb{C}$, so it is either $\mathbb{R} \oplus \mathbb{R} \oplus \mathbb{R}$ or $\mathbb{R} \oplus \mathbb{C}$. Both are commutative.

3. If $\mathcal{A}$ is a groupoid (= a category in which every morphism is an isomorphism) and $\mathcal{B}$ is a category that is equivalent to $\mathcal{A}$ then $\mathcal{B}$ is a groupoid.

Solution. TRUE. Let $F : \mathcal{B} \rightarrow \mathcal{A}$ be an equivalence. Take any morphism $f \in \text{Hom}_{\mathcal{B}}(X, Y)$. Let $g \in \text{Hom}_{\mathcal{B}}(Y, X)$ be a morphism such that Fg is inverse of Ff (inverse of Ff exists as any morphism in $\mathcal{A}$ is invertible; g exists as F is full). Then $F(f \circ g) = (Ff) \circ (Fg) = \text{id}_{FY} = F \text{id}_Y$ and $F(g \circ f) = (Fg) \circ (Ff) = \text{id}_{FX} = F \text{id}_X$. Since F is faithful this implies that $f \circ g = \text{id}_Y$ and $g \circ f = \text{id}_X$, i.e. f is an isomorphism.

4. The algebraic set $\mathcal{V}(x^2 + y^2 + z^2 - 1) \subset \mathbb{A}_{\mathbb{C}}^3$ is reducible.

Solution. FALSE. We claim that the ideal $(x^2 + y^2 + z^2 - 1) \subset \mathbb{C}[x, y, z]$ is prime. Since $\mathbb{C}[x, y, z]$ is UFD it is sufficient to show that polynomial $x^2 + y^2 + z^2 - 1 \in \mathbb{C}[x, y, z]$ is irreducible. Assume it is not, i.e. $x^2 + y^2 + z^2 - 1 = pq$ for some non-constant polynomials p, q . Then both polynomials p, q should be of degree 1, i.e. linear. By a suitable scaling we can write $p = l + 1$ and $q = m - 1$ where l, m are *homogeneous* linear polynomials. Then $pq = lm + m - l - 1$ and we deduce that $l = m$ since $x^2 + y^2 + z^2 - 1$ has no linear terms. Thus we are looking for a homogeneous linear polynomial l such that $l^2 = x^2 + y^2 + z^2$. Such polynomial does not exist: if $l = \alpha x + \beta y + \gamma z$ then l^2 involves the term $2\alpha\beta xy$, so $\alpha\beta = 0$, and l^2 misses either x^2 or y^2 (or both). This is a contradiction.

5. Let χ be the character of the symmetric group S_4 that is induced from the trivial character of the subgroup S_3 . The inner product (χ, χ) equals 2.

Solution. TRUE. Let $\mathbb{C}$ be the trivial 1-dimensional representation of S_3 . Then the induced representation $\text{Ind}_{S_3}^{S_4} \mathbb{C}$ has character χ . It is the representation corresponding to the action of the group S_4 on set of cosets $X = S_4/S_3$. This is the usual action of S_4 on the set of 4 points. Thus it is easy to compute its character. Namely, the group S_4 has 5 conjugacy classes with representatives 1, (12), (12)(34), (123), (1234); their sizes are respectively 1, 6, 3, 8, 6. For any group G acting on a set X the corresponding character is given by $\chi(g) = |\{x \in X \mid gx = x\}|$. Thus in our case we have:

$$\chi(1) = 4, \quad \chi((12)) = 2, \quad \chi((12)(34)) = 0, \quad \chi((123)) = 1, \quad \chi((1234)) = 0.$$

By definition we have $(\chi, \chi) = \frac{1}{|G|} \sum_i c_i \chi(g_i) \overline{\chi(g_i)}$ where sum is over the conjugacy classes C_i , $g_i \in C_i$, $c_i = |C_i|$. Thus we have $(\chi, \chi) = \frac{1}{24}(1 \cdot 16 + 6 \cdot 4 + 3 \cdot 0 + 8 \cdot 1 + 6 \cdot 0) = 2$.

Comment. This is a special case of a general result. Suppose that G is a finite group and X is a finite G -set with $|X| > 1$. Let χ be the character of the corresponding permutation representation $\mathbb{C}X$. Then $(1, \chi)$ is equal to the number of orbits of G on X . Hence, $(\chi, \chi) = (1, \chi^2)$ is the number of orbit of G on $X \times X$. To apply this to the problem in hand, it remains to note that S_4 has two orbits on $\{1, \dots, 4\} \times \{1, \dots, 4\}$.

Part III. Give complete answers to the following (14 points each):

1. Let F be a field of characteristic $p > 0$ and consider the polynomial

$$f(x) := x^p - x - c \in F[x]$$

for some $c \in F$. Let α be a root of $f(x)$ in an algebraic closure $\overline{F}$ of F .

- (a) Prove that $f(x)$ has roots $\alpha, \alpha + 1, \dots, \alpha + p - 1$.
- (b) Prove that $F(\alpha)/F$ is a Galois extension.

- (c) Prove that $f(x)$ is either irreducible or splits into linear factors in $F[x]$.
 (Hint: Assume that $\alpha \notin F$ and construct some elements of $\text{Gal}(F(\alpha)/F)$.)

Solution. (a) By Freshman's dream, we have

$$f(\alpha + 1) = (\alpha + 1)^p - (\alpha + 1) - a = f(\alpha) = 0.$$

So if α is a root, so is $\alpha + 1$. Thus $f(x)$ has roots $\alpha, \alpha + 1, \alpha + 2, \dots, \alpha + p - 1$. Since the degree of $f(x)$ is p this is a complete list of the roots.

(b) By (a), $F(\alpha)$ is a splitting field of $f(x)$ over F , and $f(x)$ has distinct roots in this splitting field. Hence, $F(\alpha)/F$ is normal and separable, i.e., a Galois extension.

(c) It is clear from (a) that if $\alpha \in F$, then all the roots of $f(x)$ are in F and $f(x)$ splits into linear factors. Now assume that $\alpha \notin F$. Let $m(x)$ be the minimal polynomial of α over F . It is a monic irreducible factor of $f(x)$. To complete the proof we need to show that $m(x) = f(x)$. We have that $F(\alpha) \cong F[x]/(m(x))$. The degree of $m(x)$ is ≥ 2 , so it has another root in $F(\alpha)$. By part (a) this root must be $\alpha + r$ for some $1 \leq r \leq p - 1$. Thus we have an automorphism g of $F(\alpha)/F$ sending α to $\alpha + r$. It is clear that the order of this automorphism is p . Thus the Galois group of $F(\alpha)/F$ has order $\geq p$. By the main theorem of Galois theory (applicable by part (b)) this order equals $[F(\alpha) : F] = \deg(m(x)) \leq p$. Thus $\deg(m(x)) = p$ and $m(x) = f(x)$.

2. Let R and S be rings, U be a left R -module, V be an (R, S) -bimodule and W be a left S -module.

- (a) Construct a homomorphism of Abelian groups

$$\alpha_U : \text{Hom}_R(U, V) \otimes_S W \rightarrow \text{Hom}_R(U, V \otimes_S W)$$

which is natural in U , such that α_R is an isomorphism (here, R is the left regular R -module). You should explain carefully why your homomorphism is well defined, but there is no need to justify the naturality.

- (b) Prove that α_U is an isomorphism when U is a finitely generated projective left R -module.

Solution. (a) Consider the function $\text{Hom}_R(U, V) \times W \rightarrow \text{Hom}_R(U, V \otimes_S W)$ mapping (f, w) to the function $\theta_{f,w} : u \mapsto f(u) \otimes w$. We should check that $\theta_{f,w}$ is indeed an R -module homomorphism: for $r \in R$ we have that $\theta_{f,w}(ru) = f(ru) \otimes w = rf(u) \otimes w = r\theta_{f,w}(u)$, QED.

The function just defined is S -balanced, i.e., $\theta_{fs,w} = \theta_{f,sw}$ for $s \in S$. This follows because $\theta_{fs,w}(u) = (fs)(u) \otimes w = f(u)s \otimes w = f(u) \otimes sw = \theta_{f,sw}(u)$. Hence, by the universal property of tensor product, we get induced a well-defined homomorphism of Abelian groups $\alpha_U : \text{Hom}_R(U, V) \otimes_S W \rightarrow \text{Hom}_R(U, V \otimes_S W)$ taking $f \otimes w$ to $\theta_{f,w}$.

It remains to check that α_R is an isomorphism. For this, we consider the diagram

$$\begin{array}{ccc} \mathrm{Hom}_R(R, V) \otimes_S W & \xrightarrow{\alpha_R} & \mathrm{Hom}_R(R, V \otimes_S W) \\ \downarrow \mathrm{ev}_1 \otimes \mathrm{id}_W & & \downarrow \mathrm{ev}_1 \\ V \otimes_S W & \xrightarrow{\mathrm{id}} & V \otimes_S W. \end{array}$$

The vertical maps and the bottom map are isomorphisms. Also the diagram commutes. To see this, take $f \in \mathrm{Hom}_R(R, V)$. The image of $f \otimes w$ around the southwest of the diagram is $f(1) \otimes w$. Its image around the northeast is $\theta_{f,w}(1) = f(1) \otimes w$ too. Hence, the top map is an isomorphism.

(b) If U is a finitely generated projective module, it is a summand of $R^{\oplus n}$ for some $n \geq 0$. Thus, there exist R -module homomorphisms $f : U \rightarrow R^{\oplus n}$ (the inclusion) and $g : R^{\oplus n} \rightarrow U$ (the projection) such that $g \circ f = \mathrm{id}_U$. We showed in (a) that α_R is an isomorphism. Similarly, using that $\mathrm{Hom}_R(R^{\oplus n}, V) \cong V^{\oplus n}$, the map $\alpha_{R^{\oplus n}}$ is an isomorphism. Naturality implies that the following diagram commutes:

$$\begin{array}{ccc} 0 & & 0 \\ \downarrow & & \downarrow \\ \mathrm{Hom}_R(U, V) \otimes_S W & \xrightarrow{\alpha_U} & \mathrm{Hom}_R(U, V \otimes_S W) \\ g^* \otimes \mathrm{id}_W \downarrow & & \downarrow g^* \\ \mathrm{Hom}_R(R^{\oplus n}, V) \otimes_S W & \xrightarrow{\alpha_{R^{\oplus n}}} & \mathrm{Hom}_R(R^{\oplus n}, V \otimes_S W) \\ f^* \otimes \mathrm{id}_W \downarrow & & \downarrow f^* \\ \mathrm{Hom}_R(U, V) \otimes_S W & \xrightarrow{\alpha_U} & \mathrm{Hom}_R(U, V \otimes_S W) \\ \downarrow & & \downarrow \\ 0 & & 0 \end{array}$$

where the vertical maps are induced by g and f , respectively. The composition of the vertical maps are identities, hence, the vertical diagrams are short exact sequences. The middle horizontal map is an isomorphism. From the top square, it follows that α_U is injective, and from the bottom one, it follows that α_U is surjective. Hence, α_U is an isomorphism.

3. A *non-trivial idempotent* in a commutative ring R is an element $e \in R - \{0, 1\}$ such that $e^2 = e$. Note that $e(1 - e) = 0$, hence, non-trivial idempotents are (non-zero) zero divisors.

- (a) Prove that the ring $R := \mathbb{C}[x, y]/(xy)$ contains a (non-zero) zero divisor but has no non-trivial idempotents¹.
- (b) Continue with $R = \mathbb{C}[x, y]/(xy)$ as in (a), and let S be the multiplicative set $1, f, f^2, \dots$ where $f = x + y$. Find a non-trivial idempotent in the localization $S^{-1}R = R_f$.

¹Geometrically, this means that the algebraic set $\mathcal{V}(xy)$ is *connected*.

Solution. (a) It is clear that R is spanned by the monomials $1, x, x^2, \dots, y, y^2, \dots$. In fact, these monomials are also linearly independent. This follows because the ideal (xy) has linear basis $\{x^i y^j \mid i, j \geq 1\}$. In particular, this shows that $x \neq 0$. Since $xy = 0$, it is a non-zero zero divisor.

Now suppose that $e \in R - \{0, 1\}$ is an idempotent. We have that $e = \lambda + a_1 x + a_2 x^2 + \dots + b_1 y + b_2 y^2 + \dots$. Let i be minimal such that $a_i \neq 0$. Since e^2 has constant term λ^2 and its x^i -coefficient is $2\lambda a_i$, we get that $\lambda^2 = \lambda$ and $2\lambda = 1$, which is a contradiction. Thus we have that $a_i = 0$ for all i . Similarly, $b_i = 0$ for all i . But then $e^2 = e$ implies that $\lambda^2 = \lambda$, so $e = 0$ or $e = 1$, i.e., it is trivial.

(b) As noted in the question, non-trivial idempotents are non-zero zero divisors. This is meant to suggest the idea of looking at some interesting zero divisor. One such is $\frac{x}{f}$. It is an idempotent because $\frac{x^2}{f^2} = \frac{x}{f}$. To see this, by the definition of localization, we need to check that $(x^2 f) f^n = (x f^2) f^n$ for some $n \geq 0$. In fact, this is already the case when $n = 0$: $x^2(x + y) = x^3 = x(x + y)^2$.

It remains to show that $\frac{x}{f} \neq 0$ and $\frac{x}{f} \neq 1$. To prove the former, $\frac{x}{f} = 0$ is equivalent to $x f^n = 0$ in R for some $n \geq 0$, which is not the case since $x f^n = x^{n+1} \neq 0$. Similarly, $\frac{y}{f} \neq 0$. Finally, to show that $\frac{x}{f} \neq 1$, note that $1 = \frac{x}{f} + \frac{y}{f}$, so $\frac{x}{f} = 1$ is equivalent to $\frac{y}{f} = 0$, and we just observed that this is not the case.

Comment: With a bit more effort one can show that $R_f \simeq \mathbb{C}[x, x^{-1}] \oplus \mathbb{C}[y, y^{-1}]$.